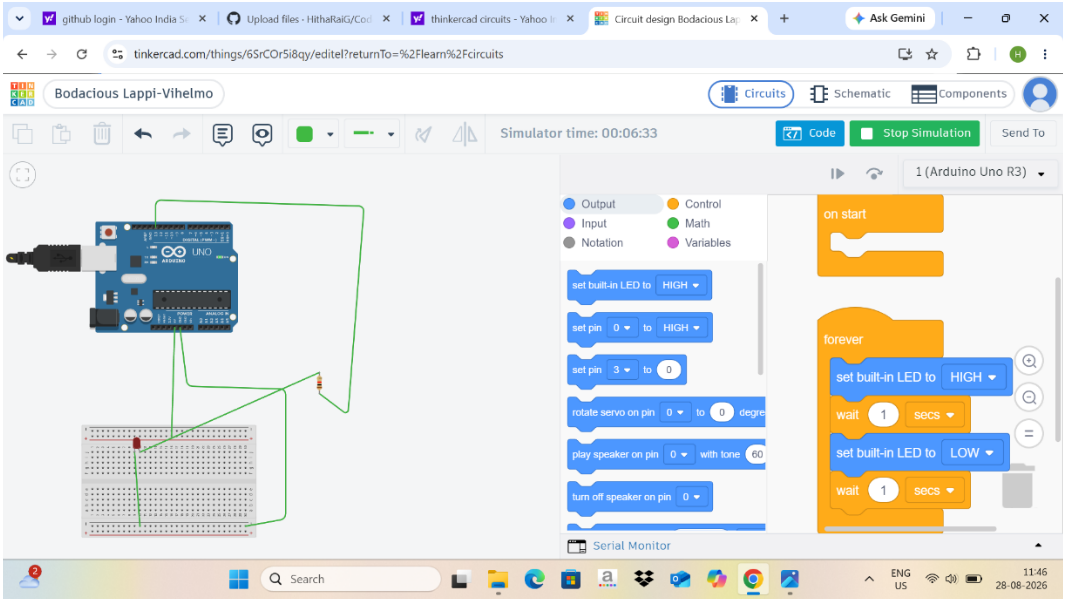
SENSOR-BASED SIMULATION

LED On/Off Using a Sensor



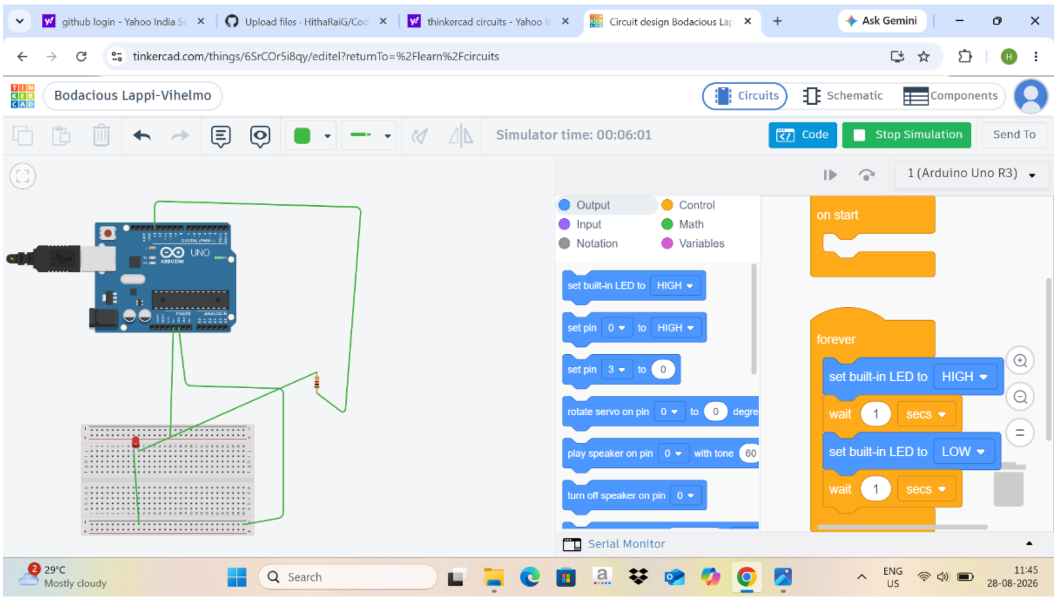


Fig : Simulation of LED

The block code shown corresponds to turning the LED ON and OFF every 1 second.

1.Arduino code

int LED = 13;

void setup()

```
{
  pinMode(LED, OUTPUT);
}
void loop()
{
  digitalWrite(LED, HIGH);
  delay(1000);
  digitalWrite(LED, LOW);
  delay(1000);
}
```

2.Explanation

1. int LED = 13;

   We assign Arduino digital pin 13 to the LED.

2. void setup()

   - This part runs only once when the Arduino starts.

     pinMode(LED, OUTPUT);

   - It configures pin 13 as an output pin.

3. void loop()

   - This part runs continuously.

     digitalWrite(LED, HIGH);

   - Turns the LED ON.

     delay(1000);

   - Waits for 1000 milliseconds = 1 second.

     digitalWrite(LED, LOW);

   - Turns the LED OFF.

     delay(1000);

   - Waits another 1 second.
   - Then loop() starts again

3.Working

LED ON → 1 second → LED OFF → 1 second → repeat continuously

Components used

1. Arduino UNO
2. LED
3. Resistor (typically 220 Ω)
4. Breadboard
5. Jumper wires